

Indian Statistical Institute
B. Statistical Data Science II (2025-26)
Assignment - 2026

Paper: **Mathematics IV**

Deadline: **03/05/2026**

Date: **13/04/2026**

Full Marks : **10**

Note: Symbols used have their usual meaning.

Write your Name, Roll Number, Centre Name, and Email ID clearly at the top of your assignment (answer sheets). Compile your work into a single PDF file—either by scanning handwritten pages or generating it from a LaTeX/Word document. Save the file using the format: “**Your Name_Roll No.pdf**” and submit it through Google Form by **03/05/2026**.

Link for the **Google Form** is <https://forms.gle/gmo6o8A1X2ioUSi8A>.

Part-I: Ordinary Differential Equations

1. (a) Solve the following two equations

i. $\frac{dy}{dx} = \frac{x + 4y - 2}{4x - y + 9}$.

ii. $\frac{dy}{dx} = \frac{6x + 4y + 1}{3x + 2y + 1}$.

- (b) Find the orthogonal trajectories of the family of confocal conics.

$$\frac{x^2}{a^2 + \lambda} + \frac{y^2}{b^2 + \lambda} = 1 \quad (\lambda = \text{parameter}).$$

2. (a) Let $D \subset \mathbb{R}^2$ be a closed rectangular domain and $f : D \rightarrow \mathbb{R}$ is a continuous function. Suppose, $T(y)$ be a function given by,

$$T(y)(x) = y_0 + \int_{x_0}^x f(t, y(t)) dt,$$

where $y(x_0) = y_0$ and $(x_0, y_0) \in D$. Show that there exist at least one solution for $T(y) = y$. Is $T(y)$ necessarily a *contraction map*¹? Justify your answer.

- (b) Consider the differential equation

$$a_0(x) \frac{d^2 y}{dx^2} + a_1(x) \frac{dy}{dx} + a_2(x) y = 0, \tag{1}$$

where a_0, a_1 and a_2 are continuous for all x , $-\infty < x < \infty$, and $a_0(x) \neq 0$ for all values of x .

¹A contraction map on a metric space X is a function $T : X \rightarrow X$ such that for some $\alpha < 1$

$$d(T(u) - T(v)) \leq \alpha d(u, v),$$

for all $x, y \in X$.

- i. Let f be a nontrivial solution of the differential equation (1), let f' denote the derivative of f , and let $x_0 \in [a, b]$. Prove that if $f(x_0) = 0$, then $f'(x_0) \neq 0$.
- ii. Let f and g be two distinct nontrivial solutions of the differential equation (1), and suppose there exists $x_0 \in [a, b]$ such that $f(x_0) = g(x_0) = 0$. Prove that there exists a constant c such that $f = cg$.

3. (a) Find the solution of the following linear system

$$\frac{d\mathbf{x}}{dt} = \begin{pmatrix} 8 & 12 & -2 \\ -3 & -4 & 1 \\ -1 & -2 & 2 \end{pmatrix} \mathbf{x}.$$

- (b) Find the solution of the nonhomogeneous linear system

$$\frac{d\mathbf{x}}{dt} = \begin{pmatrix} -1 & 1 \\ -12 & 6 \end{pmatrix} \mathbf{x} + \begin{pmatrix} 3e^{4t} \\ 8e^{4t} \end{pmatrix}$$

that satisfies the initial condition $\Phi(0) = \begin{pmatrix} 2 \\ 4 \end{pmatrix}$.

4. (a) Solve the following linear system of differential equations using Laplace transformation²

$$\begin{aligned} \frac{d^2x}{dt^2} - 3\frac{dx}{dt} + \frac{dy}{dt} + 2x - y &= 0 \\ \frac{dx}{dt} + \frac{dy}{dt} - 2x + y &= 0 \end{aligned}$$

with the given initial condition $x(0) = 0, y(0) = -1, \dot{x}(0) = 0$.

- (b) Let

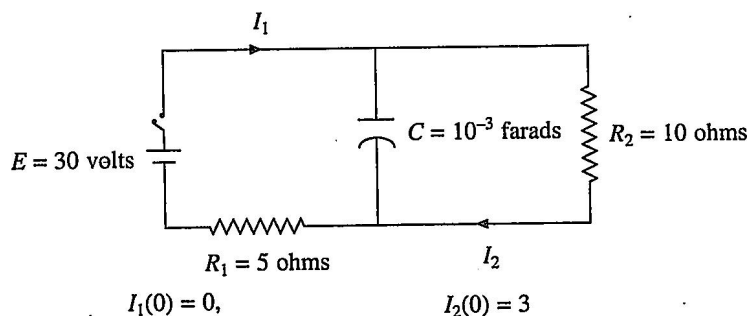
$$\frac{d^2x}{dt^2} + [q(t) + k^2]x = 0, \tag{2}$$

where q is a continuous real valued function and k is a real number. Show that

- i. if $q(t) > 0$ for $t > 0$ and k is a positive constant, then every real solution of (2) has an infinite number of positive zeros.
- ii. if $k = 0$ and $q(t) < 0$ throughout $a \leq t \leq b$, then no nontrivial solution of (2) can have more than one zero on $a \leq t \leq b$.

5. (a) Analyze the following electric circuits to obtain currents in each segment and find the steady state currents I_1 and I_2 at time t .

²The Laplace transformation $\mathcal{L}(f)$ of a real valued function f is defined by $\mathcal{L}(f) = \int_0^\infty e^{-st} f(t) dt$ [= $F(s)$, say].



Here, E denotes electromotive force, C denotes Capacitor and R_i ($i=1,2$) denotes resistor.

- (b) A coil spring having spring constant 20 lb/ft is suspended from the ceiling. A 32-lb weight is attached to the lower end of the spring and comes to rest in its equilibrium position. Beginning at $t = 0$ an external force given by

$$F(t) = 40 \cos 2t$$

is applied to the system. This force then remains in effect until $t = \pi$, at which instant it ceases to be applied. For $t > \pi$, no external forces are present. The medium offers a resistance in pounds numerically equal to $4 \left(\frac{dx}{dt} \right)$, where $\frac{dx}{dt}$ is the instantaneous velocity in feet per second. Find the displacement of the weight as a function of time for all $t \geq 0$.

Part-II: Calculus

6. (a) Consider the two lines given by the parametric equations, Line 1 : $X = x_0 + x_1t, Y = y_0 + y_1t, Z = z_0 + z_1t, t \in \mathbb{R}$; and Line 2 : $X = x_2 + x_3s, Y = y_2 + y_3s, Z = z_2 + z_3s, s \in \mathbb{R}$. In terms of $x_i, y_i, z_i, 0 \leq i \leq 3$, find an equivalent condition for the existence of a plane in $\mathbb{R}^3$ containing both of these two lines.
- (b) Find the length of the shortest distance between the two non-intersecting lines $\mathbf{r} = t\mathbf{b}$ and $\mathbf{r} = \mathbf{a} + s\mathbf{c}$. Also determine the equation of the line which cuts both the lines at right angles.
7. (a) Let P be a point in $\mathbb{R}^3$ that is not contained in the plane Π that passes through the three non-collinear points A, B and C . Show that the distance between P and Π is given by

$$\frac{|\mathbf{p} \cdot (\mathbf{b} \times \mathbf{c})|}{\|\mathbf{b} \times \mathbf{c}\|},$$

where $\mathbf{p} = \vec{AP}, \mathbf{b} = \vec{AB}, \mathbf{c} = \vec{AC}$.

- (b) Prove or disprove that— “if $\mathbf{b}, \mathbf{c}$ and $\mathbf{d}$ are non-coplanar, then $(\mathbf{a} \times \mathbf{b}) \times (\mathbf{c} \times \mathbf{d}) + (\mathbf{a} \times \mathbf{c}) \times (\mathbf{d} \times \mathbf{b}) + (\mathbf{a} \times \mathbf{d}) \times (\mathbf{b} \times \mathbf{c})$ is parallel to $\mathbf{a}$ ”.

8. (a) Let $(x_0, y_0, z_0) \in \mathbb{R}^3$. Consider the sphere in $\mathbb{R}^3$ defined by the equation,

$$(X - x_0)^2 + (Y - y_0)^2 + (Z - z_0)^2 = R^2, \text{ for some constant } R;$$

Let (x_1, y_1, z_1) be a point on this sphere. Determine the equation of the tangent plane to the sphere at the point (x_1, y_1, z_1) .

- (b) Prove the following statement, or disprove it using a counterexample and an explanation: Consider a smooth map $f : \mathbb{R}^m \rightarrow \mathbb{R}^n$. Let f' denote the derivative of f , which is considered as a function on $\mathbb{R}^m$. Thus, f' is a map from $\mathbb{R}^m$ to $M_{m \times n}(\mathbb{R}) \simeq \mathbb{R}^{mn}$, where $M_{m \times n}(\mathbb{R})$ is the set of all $m \times n$ matrices over $\mathbb{R}$. As derivative of any linear map is the same linear map itself, the double derivative of f , denoted f'' , must be same map as f' .

9. (a) Prove that $\nabla^2 f(r) = \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{df}{dr} \right)$. Hence find $f(r)$ such that $\nabla^2 f(r) = 0$.

- (b) Prove or disprove the following statement:

“Under a rotation of coordinate axes, the Laplacian operator $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$ is invariant”.

10. (a) Evaluate the surface integral $\iint_S (x^2 + y^2) dS$, where S is the surface of the cone $z^2 = 3(x^2 + y^2)$ bounded by $z = 0$ and $z = 3$.

- (b) A billiard ball on a pool table follows the path $c : [-1, 1] \rightarrow \mathbb{R}^2$, defined by

$$c(t) = (x(t), y(t)) = \left(t, \left| t - \frac{1}{2} \right| \right).$$

Find the distance traveled by the ball.